The Characteristic Equation

Geometric Algorithms

CAS CS 132

Objectives

- Briefly recap eigenvalues and eigenvectors
- Get a primer on **determinants**
- Determine how to **find eigenvalues** (not just verify them)

Keyword

eigenvectors

eigenvalues

eigenspaces

eigenbases

determinant

characteristic equation

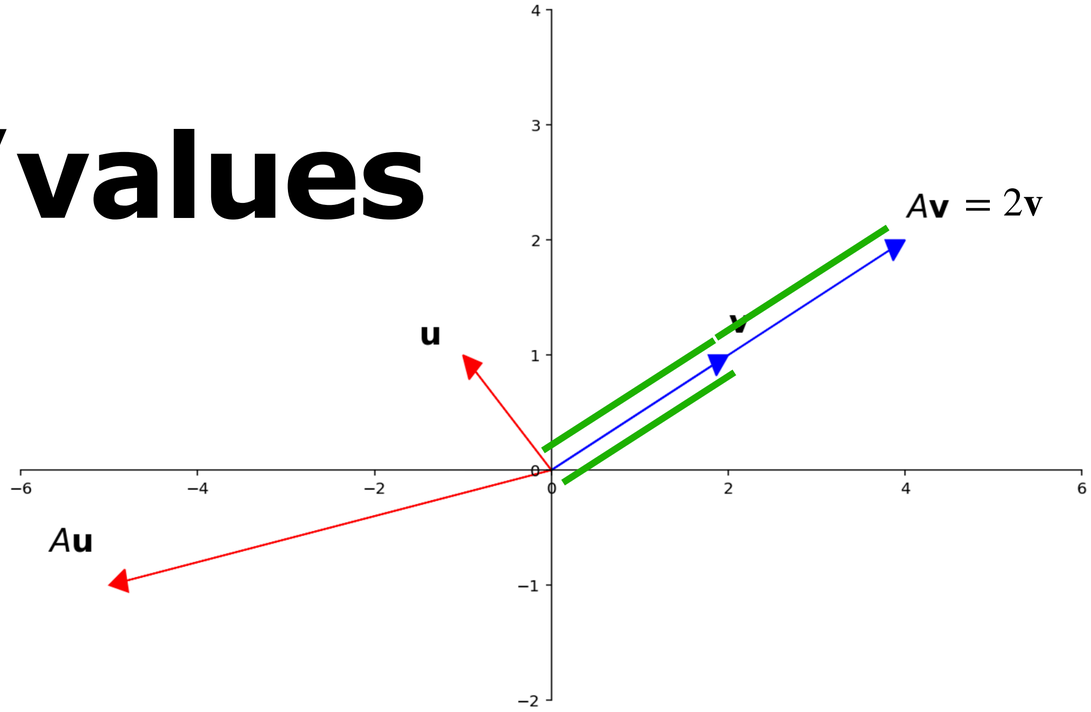
polynomial roots

triangular matrices

multiplicity

Recap

Recall: Eigenvectors/values



Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector and eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if

$$A\mathbf{v} = \lambda\mathbf{v}$$

Eigenvectors are just scaled by A

Recall: Verifying Eigenvectors

Q. Determine if $\mathbf{v}$ is an eigenvector of A and determine the corresponding eigenvalues

A. Easy. Work out the matrix-vector multiplication

Ex.

corresponding eigenvalue

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 13 \\ 9 \end{bmatrix}$$



Recall: Verifying Eigenvalues

Q. Find an eigenvector of A whose eigenvalue is λ

A. Find a nontrivial solution to

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$



If we don't need the vector we can just show that $A - \lambda I$ *is* ***not*** *invertible (by IMT)*

Recall: Finding Eigenspaces

Q. Find a basis for the eigenspace of A for λ

A. Find a basis for $\text{Nul}(A - \lambda I)$

(we did this for our recap problem)

Practice Problem

$$A = \begin{bmatrix} 5 & 2 & 3 & 0 \\ -1 & 2 & -3 & 1 \\ 2 & 4 & 10 & 0 \\ 1 & 2 & 3 & 5 \end{bmatrix}$$

Determine the dimension of the eigenspace of A for the eigenvalue 4

(try not to do any row reductions)

Answer

$$A = \begin{bmatrix} 5 & 2 & 3 & 0 \\ -1 & 2 & -3 & 1 \\ 2 & 4 & 10 & 0 \\ 1 & 2 & 3 & 5 \end{bmatrix}$$

$\dim(\mathrm{Nul}(A-4I)) = 2$

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ -1 & -2 & -3 & 1 \\ 2 & 4 & 6 & 0 \\ 1 & 2 & 3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$v_1 \quad v_2 \quad v_3$

$\dim(\mathrm{Col}(A-4I)) + \dim(\mathrm{Nul}(A-4I)) = 4$

$\dim(\mathrm{Nul}(A-4I)) = 2$

2 as v_1, v_2, v_3 lin. dep.

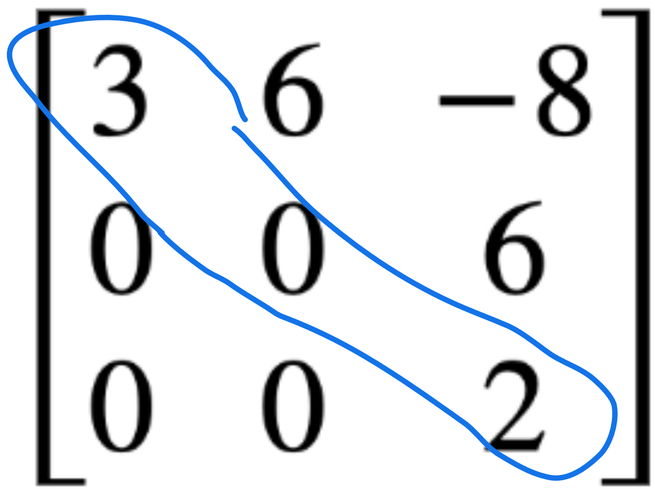
Recall: Eigenvectors and Linear Dynamical Systems

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if $\mathbf{v}_0$ can be written in terms of eigenvectors $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k$ of A with eigenvalues $|\lambda_1| > |\lambda_2| \ldots \geq |\lambda_k|$ then $\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$ for some constant c_1

$$\frac{\vec{v}_k}{(\lambda_1^k)} \xrightarrow{k \to \infty} c_1 \vec{b}_1$$

Eigenvalues of Triangular Matrices

Thm. The eigenvalues of a triangular matrix are its entries along the diagonal



Determinants

What kind of thing is the determinant?

A determinant, written $\det(A)$, is a *number* associated with a matrix

In broad strokes, it's a big sum of products of entries of A

A Scary-Looking Definition (we won't use)

$$\det(A) = \sum_{\sigma \in S_n} (-1)^{\text{sgn}(\sigma)} A_{1\sigma(1)} A_{2\sigma(2)} \cdots A_{n\sigma(n)}$$

We can think of this as a procedure

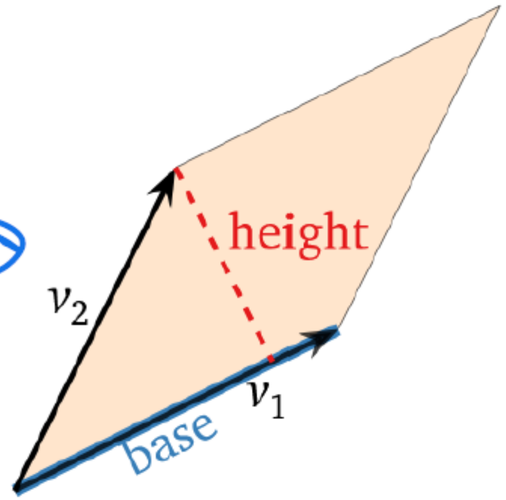
```
1 FUNCTION det(A):
2   total = 0
3   FOR all matrix B we can get by swapping a bunch of rows of A:
4     s = 1 IF (# of swaps necessary) is even ELSE -1
5     total += s * (product of the diagonal entries of B)
6   RETURN total
```

A Geometric Interpretation: Volume

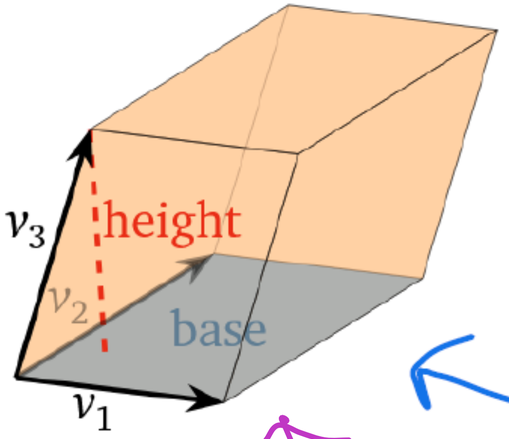
$|\det[\vec{v}_1\ \vec{v}_2]|$

$=$

$\text{vol}(P)$



2-dim'l vol is area



$|\det[\vec{v}_1\ \vec{v}_2\ \vec{v}_3]|$

$=$

$\text{vol}(P)$

The Determinant of 2×2 Matrices

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$(-1)^0 ad$ is the product of the diagonal

$(-1)^1 bc$ is the product of the diagonal after 1 swap

The Determinant of 2×2 Matrices

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow^{0} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$(-1)^0 ad$$

The Determinant of 2×2 Matrices

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow^{1} \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

$$(-1)^{1} cb$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{0} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$(-1)^{0} aei$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{2} \begin{bmatrix} g & h & i \\ a & b & c \\ d & e & f \end{bmatrix}$$

$$(-1)^2 gbf$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

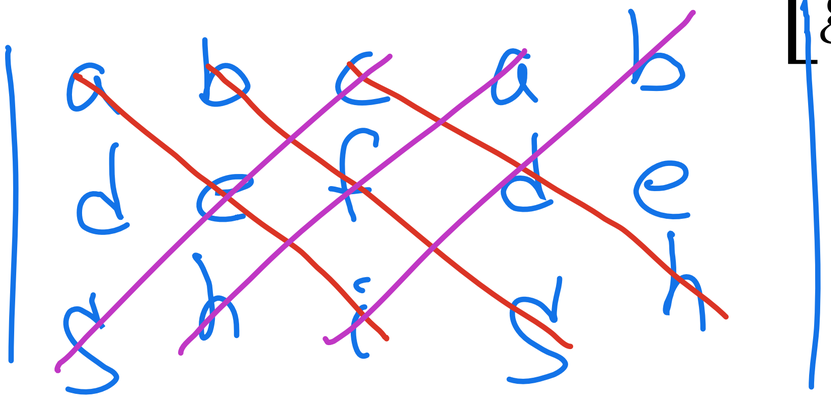
$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{2} \begin{bmatrix} d & e & f \\ g & h & i \\ a & b & c \end{bmatrix}$$

$$(-1)^2 dhc$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

Sarras' rule



$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{1} \begin{bmatrix} g & h & i \\ d & e & f \\ a & b & c \end{bmatrix}$$

does $(-1)^1 gec$

NOT extend to other dim'ns

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{1} \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix}$$

$$(-1)^{1} dbi$$

The Determinant of 3×3 matrices

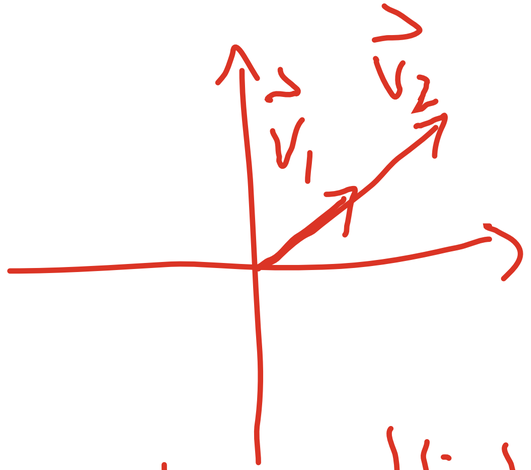
$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{1} \begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix}$$

$$(-1)^{1} ahf$$

What if columns of a matrix are lin. dep.?

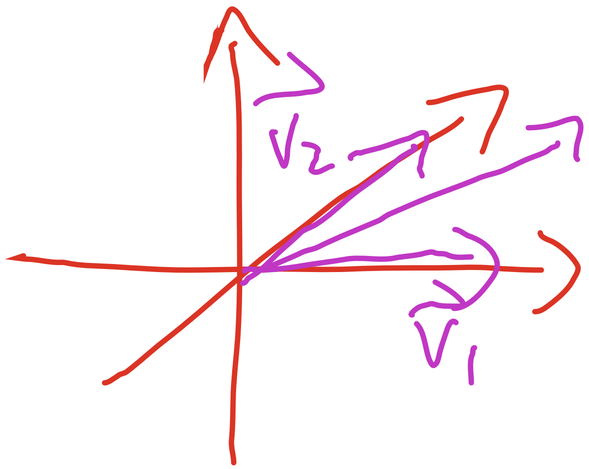
In 2D:



scalar multiples

$$\text{vol}(P) = 0 = \det\left(\left[\vec{v}_1\ \vec{v}_2\right]\right)$$

In 3D:



$\vec{v}_3$ in span$(\{\vec{v}_1, \vec{v}_2\})$

$$\text{vol}(P) = 0 = \det\left(\left[\vec{v}_1\ \vec{v}_2\ \vec{v}_3\right]\right)$$

Determinants and Invertibility

Thm. A is invertible if and only if $\det(A) \neq 0$

So we can extend the IMT:

- A is invertible
- $\det(A) \neq 0$
- 0 is not an eigenvalue

(these must be all true or all false)

Determinants (the definition we'll use)

$$\det(A) = \frac{(-1)^s}{c} U_{11}U_{22}\cdots U_{nn}$$

product of diagonal entries

0 if A is not invertible

Df. The **determinant** of a matrix A is given by the above equation, where

» U is an *echelon form* of A

» s is the number of row *swaps* used to get U

» c is the *product of all scalings* used to get U

(DOES NOT INCLUDE ROW ADD'N OPS)

Example

$$\begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$s = 1$
$c = 1$

$$\sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix} \xrightarrow[\text{swap}]{R_2 R_3} \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 1/3 \end{bmatrix}$$

$s = 0$
$c = 1$

$$\det(A) = \frac{(-1)^1}{1} 2 = -2$$

$$\frac{(-1)^0}{1}(-2) = -2$$

The definition holds no matter which row operations you use

How To: Determinants

Q. Determine the determinant of a matrix A

A.

1. Convert A to an echelon form U
2. Keep track of the number of row swaps you used, call this s, and the product of all scalings, call this c
3. Determine the product of entries along the diagonal of U, call this P
4. The determinant of A is $\frac{(-1)^s P}{c}$

Another way: cofactor expansion

(Useful when certain rows/columns are sparse)

Expansion along row i:

$$\det(B) = \sum_{j=1}^{n} (-1)^{i+j} b_{i,j} m_{i,j},$$

$m_{i,j}$ ← det of submatrix formed by removing row i & col. j

$(-1)^{i+j}$ ↑

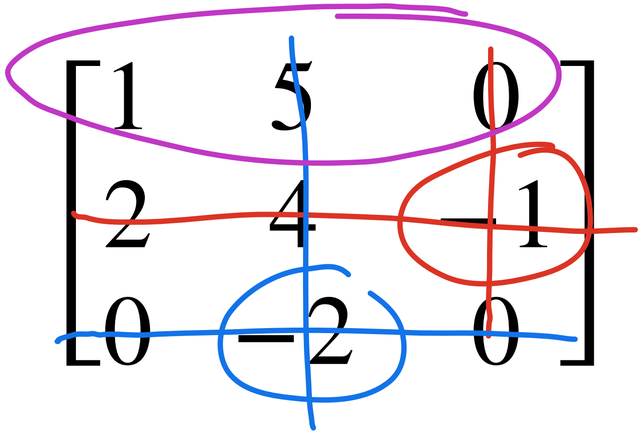
$b_{i,j}$ ↑ entries of B

Expansion along column j:

$$\det(B) = \sum_{i=1}^{n} (-1)^{i+j} b_{i,j} m_{i,j}.$$

(notation stolen from "Laplace expansion" Wikipedia page)

Example (Again)



expansion along col 3:

$$(-1)^{2+3}(-1)\begin{vmatrix} 1 & 5 \\ 0 & -2 \end{vmatrix} = (-1)(-1)(-2) = -2$$

row 3

$$(-1)^{3+2}(-2)\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} = (-1)(-2)(-1) = -2$$

The Shorter Version

numpy.linalg.det(A)

Beyond small matrices, we'll just use a computer

Properties of Determinants

Properties of Determinants (1)

$$\det(AB) = \det(A)\det(B)$$

Cor. AB is invertible $\Leftrightarrow$ A and B are invertible

$\Updownarrow$

$\det(AB) \neq 0$

$\Updownarrow$

$\det(A)\det(B) \neq 0$

Question

Use the fact that $\det(AB) = \det(A)\det(B)$ *to give an expression for* $\det(A^{-1})$ *in terms of* $\det(A)$

Hint. What is $\det(I)$*?*

$\det(AA^{-1}) = \det(I) = 1$

$\det(A)\det(A^{-1})$

$$\Rightarrow \det(A^{-1}) = \frac{1}{\det(A)}$$

Properties of Determinants (2)

$$\det(A^T) = \det(A)$$

It follows that A^T is invertible if and only if A is invertible

(we also won't verify this)

Question

orthogonal matrices

If $A^{-1} = A^T$*, then what is* $\det(A)$*?*

$$\det(A^{-1}) = \det(A^T) = \det(A) = \frac{1}{\det(A)}$$

$$\det(A^{-1}) = \frac{1}{\det A}$$

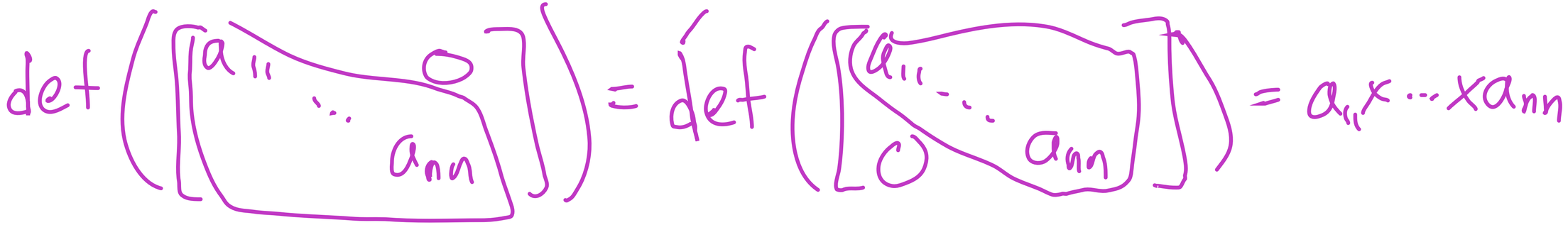
$$\Rightarrow (\det(A))^2 = 1$$

$$\det(A) = \pm 1$$

Properties of Determinants (3)

Thm. If A is triangular, then $\det(A)$ is the product of entries along the diagonal





Question

$$\begin{bmatrix} 1 & 5 & -4 \\ -1 & -5 & 5 \\ -2 & -8 & 7 \end{bmatrix}$$

Find the determinant of the above matrix

Answer

$s = 1$

$c = 1$

$$\begin{bmatrix} 1 & 5 & -4 \\ -1 & -5 & 5 \\ -2 & -8 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -4 \\ 0 & 0 & 1 \\ 0 & 2 & -2 \end{bmatrix} \xrightarrow[R_2, R_3]{\text{swap}} \begin{bmatrix} 1 & 5 & -4 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det(A) = \frac{(-1)^1}{1}(2) = -2$$

$$\begin{vmatrix} 1 & 5 & -4 & 1 & 5 \\ -1 & -5 & 5 & -1 & -5 \\ -2 & -8 & 7 & -2 & -8 \end{vmatrix} = -35 - 50 - 32 + 40 + 40 + 35$$

$$= -2$$

Characteristic Equation

Finding Eigenvalues

Q. Determine the eigenvalues of A, along with their associated eigenspaces

A. Can we somehow "solve for λ" in the equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

want nonzero $\vec{x}$ that satisfy

$\text{Nul}(A - \lambda I) \neq \{\vec{0}\}$

$A - \lambda I$ not invertible

$\det(A - \lambda I) = 0$ ← char. polynomial

What kind of thing is the determinant, really?

The determinant of a matrix A is an *arithmetic expression* written in terms of the entries of A

But a matrix may not have numbers as entries

We might think of the matrix $A - \lambda I$ has having *polynomials* as entries

$$\begin{matrix} a_{11}-\lambda & a_{12} \\ a_{21} & a_{22}-\lambda \end{matrix} \quad \cdots$$

Then $\det(A - \lambda I)$ is a **polynomial**

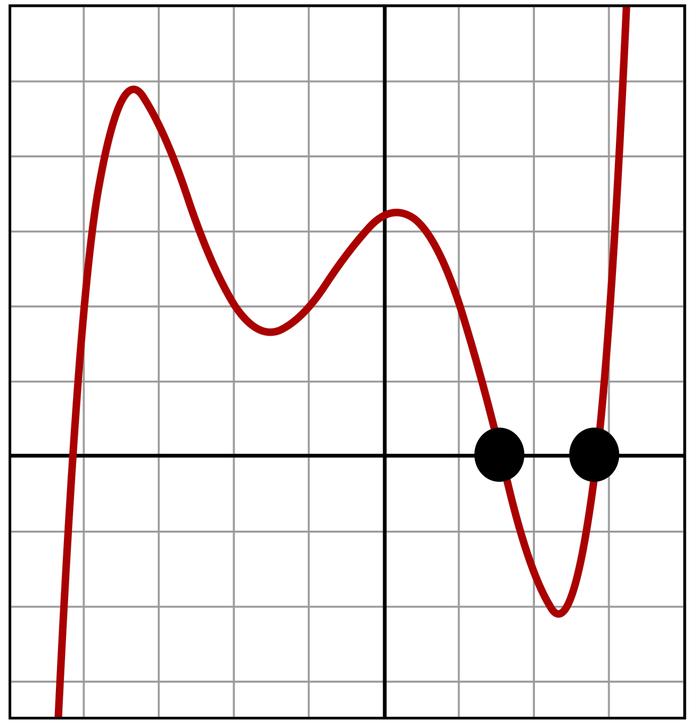
Reminder: Polynomial Roots

A **root** of a polynomial $p(x)$ is a value r such that $p(r) = 0$

(A polynomial may have many roots)

If r is a root of $p(x)$, then it is possible to find a polynomial $q(x)$ such that

$$p(x) = (x - r)q(x)$$



https://commons.wikimedia.org/wiki/File:Quintic_polynomial.svg

Characteristic Polynomial

Df. The **characteristic polynomial** of a matrix A is $\det(A - \lambda I)$ viewed as a polynomial in the *variable* λ.

This is a polynomial with the eigenvalues of A as roots

So we can "solve" for the eigenvalues in the equation

$$\det(A - \lambda I) = 0$$

Example: 2×2 Matrix

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \Rightarrow \det(A - \lambda I) = (1-\lambda)(-\lambda) - 1$$

$$= \lambda^2 - \lambda - 1$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a\lambda^2 + b\lambda + c = 0$$

$$\lambda = \frac{1 \pm \sqrt{1 - 4(1)(-1)}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

A Special Linear Dynamical System

$$\mathbf{v}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_k \qquad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{v}_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

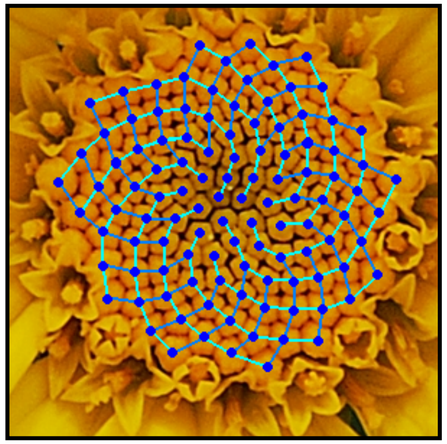
Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_k = F_{k-1} + F_{k-2}$$

```
define fib(n):
  curr, next ← 0, 1
  repeat n times:
    curr, next ← next, curr + next
  return curr
```

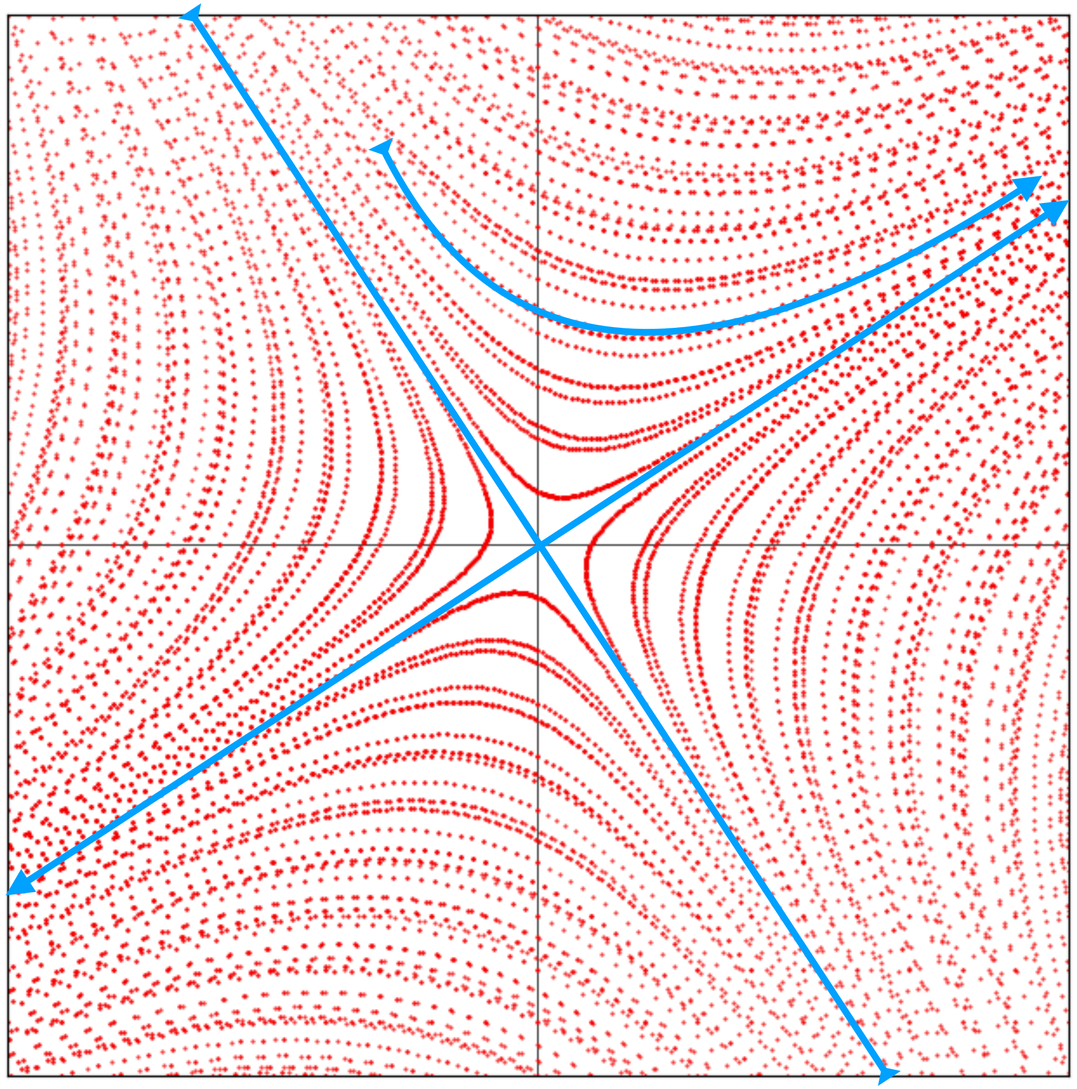


The Fibonacci numbers are defined in terms of a recurrence relation

They seem to crop-up in nature, engineering, etc.

https://commons.wikimedia.org/wiki/File:FibonacciChamomile.PNG

Recall: The Picture



$$\frac{1+\sqrt{5}}{2}$$

The eigenvalue of this matrix is the golden ratio

Golden Ratio

$$\varphi = \frac{1+\sqrt{5}}{2}$$

$$\frac{F_{k+1}}{F_k} \to \varphi \text{ as } k \to \infty$$

$$F_k \sim c\varphi^k$$

$\lambda_1 = \frac{1+\sqrt{5}}{2}$

This is the largest eigenvalue of $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ $\vec{v}_1 = \begin{bmatrix} \varphi \\ 1 \end{bmatrix}$

The "long term behavior" is the Fibonacci sequence is defined by the golden ratio

Example: Triangular matrix

$$\begin{bmatrix} 1 & -3 & 0 & 6 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & -3 & 0 & 6 \\ & -\lambda & 1 & 1 \\ & & 1-\lambda & 2 \\ 0 & & & 4-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)^2(-\lambda)(4-\lambda)$$

How To: Finding Eigenvalues

Q. Find all eigenvalues of the matrix A

A. Find the roots of the characteristic polynomial of A

An Observation: Multiplicity

$$\lambda^1(\lambda-1)^2(\lambda-4)^1$$ multiplicities

In the examples so far, we've seen a number appear as a root multiple times

This is called the **algebraic multiplicity** of the root

Is the algebraic multiplicity meaningful?

Multiplicity and Dimension

Thm. The dimension of the eigenspace of A for the eigenvalue λ is *at most* the algebraic multiplicity of λ in $\det(A - \lambda I)$

The algebraic multiplicity is an upper bound on "how large" the eigenspace is

Example

Let $A \in \mathbb{R}^{5\times5}$ with characteristic polynomial

$$(x-1)^3(x-3)(x+5)$$

» What is $\text{rank}(A)$?

» What is the minimum possible rank of $A - I$?